

Exam date: _____

Duration: 2 hours

Weighting: 40%

Calculator: Permitted

Tutor: Josh Aaron

Student name: _____

ID: _____

Instructions

- **Section A:** Answer any **10** of the 12 questions. Each is worth 4 marks. **(40 marks)**
- **Section B:** Answer any **4** of the 6 questions. Each is worth 20 marks. **(80 marks)**
- A formula sheet is attached. All necessary formulae are provided there.
- Show all working. A correct final answer with no working may not receive full marks.
- In Section A, clearly cross out any question you do not wish to be marked.

Marking summary (examiner use only)

Section A – mark the 10 questions being counted:

Q	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12		
Max	4	4	4	4	4	4	4	4	4	4	4	4	Sec A	
Score													Max	40

Section B – mark the 4 questions being counted:

Question	B1	B2	B3	B4	B5	B6		
Max marks	20	20	20	20	20	20	Sec B	
Score							Max	80

GRAND TOTAL	/120
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Section A – Answer any 10 of 12 questions (4 marks each, 40 marks total)

A1. Two friends are comparing which apps they have on their phones.

Nia's phone: Calculator, Camera, Clock, Maps, Music, Notes

Pat's phone: Calculator, Calendar, Clock, News, Photos

- (a) Which apps do **both** Nia and Pat have? *(2 marks)*
- (b) List every app that appears on **at least one** of the two phones. *(1 mark)*
- (c) How many apps does Nia have that Pat does **not** have? *(1 mark)*

A2. Let $f(x) = 3x - 2$.

- (a) Find $f(5)$. *(1 mark)*
- (b) Find $f^{-1}(x)$, the reverse (inverse) of f . *(2 marks)*
- (c) Check your answer: show that $f(f^{-1}(7)) = 7$. *(1 mark)*

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A1	/4
A2	/4
Page total	/8

A3. Consider $f(x) = x^2 - x - 6$.

- (a) Write $f(x)$ as a product of two brackets. *(2 marks)*
- (b) State the two values of x where the graph crosses the x -axis. *(1 mark)*
- (c) On a rough sketch, mark the two crossings and the y -intercept. *(1 mark)*

A4. Solve the following pair of equations and find the values of x and y .

$$2x + y = 8 \qquad x - y = 1$$

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A3	/4
A4	/4
Page total	/8

A5. Two points $A(1, 2)$ and $B(4, 8)$ lie on the coordinate plane.

- (a) Find the slope of the line through A and B . (1 mark)
- (b) Write the equation of the line through A and B . (1 mark)
- (c) A second line passes through B and is **perpendicular** to AB . Find its equation. (2 marks)

A6. Points $P(-1, 3)$ and $Q(5, -1)$ are given.

- (a) Find the midpoint M of PQ . (1 mark)
- (b) Find the distance $|PQ|$. Leave your answer in surd form if it does not simplify to a whole number. (2 marks)
- (c) Find the distance from M to P . How does it compare to $|PQ|$? (1 mark)

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A5	/4
A6	/4
Page total	/8

A7. A right-angled triangle whose three sides are all whole numbers is called a *Pythagorean triple*. The 3-4-5 triangle is the simplest example, since $3^2 + 4^2 = 9 + 16 = 25 = 5^2$.

Find **two more** right-angled triangles where every side is a whole number. For each one, write down the three lengths and show that the Pythagorean theorem holds. (2 marks) each

A8. In a right-angled triangle, the side opposite angle θ has length $\sqrt{3}$ and the hypotenuse has length 2.

- (a) Write down $\sin \theta$ and $\cos \theta$. Leave your answers in surd form. (2 marks)
- (b) Find the length of the third side (adjacent to θ). (1 mark)
- (c) Use your answers to find $\tan \theta$. (1 mark)

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A7	/4
A8	/4
Page total	/8

- A9.** (a) A 4-digit PIN uses digits 0–9 and digits **can** be repeated. How many different PINs are there? *(1 mark)*
(b) One PIN is chosen at random. What is the probability it is exactly “1234”? *(1 mark)*
(c) How many PINs are there where **no digit is repeated**? *(2 marks)*

A10. The English alphabet has 26 letters: **6 vowels** (a, e, i, o, u, y) and **20 consonants**. A letter is chosen at random from the alphabet.

- (a) What is the probability that the chosen letter is a vowel? *(1 mark)*
(b) What is the probability that the chosen letter is **not** a vowel? *(1 mark)*
(c) Two letters are chosen one after the other, **without replacement**. What is the probability that **both** are vowels? Show your working. *(2 marks)*

Marks this page	
A9	/4
A10	/4
Page total	/8

A11. A class of 20 students sat a short quiz. Their scores out of 5 are recorded below.

Score	1	2	3	4	5
Number of students	3	5	7	3	2

- (a) What score appeared most often? *(1 mark)*
 (b) Calculate the mean score. *(2 marks)*
 (c) A student is chosen at random. What is the probability their score is 4 or higher? *(1 mark)*

A12. Consider the function $f(x) = \frac{2x+4}{x}$.

- (a) Calculate $f(1)$, $f(10)$, $f(100)$, and $f(1000)$. *(2 marks)*
 (b) As x grows larger and larger without end, what single value does $f(x)$ appear to be getting closer and closer to? *(1 mark)*
 (c) What happens to $f(x)$ as x gets very close to 0? Give a brief reason. *(1 mark)*

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A11	/4
A12	/4
Page total	/8

Section B – Answer any 4 of 6 questions (20 marks each, 80 marks total)

Question B1 Quadratic Functions and Derivatives [20 marks]

Let $f(x) = x^2 - 6x + 8$.

- (a) Find the two roots of $f(x) = 0$ and the y -intercept. Use the roots to find the x -coordinate of the vertex, then calculate the y -value there. State the coordinates of the vertex. *(5 marks)*
- (b) Sketch the graph of $f(x)$, clearly showing the vertex, both roots, and the y -intercept. Label all four points with their coordinates. You do not need graph paper. *(6 marks)*
- (c) Find $f'(x)$. Use it to state the value of x at which f has its minimum, the minimum value of f , and the intervals where f is decreasing and where it is increasing. *(5 marks)*
- (d) Now consider $g(x) = f(x + 1)$. Substitute $(x + 1)$ into your expression for f and simplify $g(x)$ fully. Then describe in one sentence how the graph of g is related to the graph of f . *(4 marks)*

Marks this page	
(a)	/5
(b)	/6
(c)	/5
(d)	/4
Page total	/20

Question B2 Cubic Functions**[20 marks]**Let $g(x) = x^3 - 3x$.

- (a) Complete the table of values below.
- (3 marks)

x	-2	-1	0	1	2
$g(x)$					

- (b) Using your table, sketch the graph of $g(x)$. Mark approximately where the graph crosses the x -axis. You do not need exact values for the crossings. (4 marks)
- (c) Find $g'(x)$ by differentiating. (2 marks)
- (d) Set $g'(x) = 0$ and solve to find the x -coordinates of both turning points. Then calculate the y -value at each one. (4 marks)
- (e) For each turning point, explain whether it is a local maximum or a local minimum. Then state the full intervals over which g is increasing and over which g is decreasing. (4 marks)
- (f) As x grows to be a very large positive number, what happens to $g(x)$? What about when x becomes a very large negative number? A brief description is enough. (2 marks)
- (g) Calculate $g'(0)$. In one sentence, explain what this value tells you about the shape of the curve at $x = 0$. (1 mark)

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(a)	/3
(b)	/4
(c)	/2
(d)	/4
(e)	/4
(f)	/2
(g)	/1
Page total	/20

Question B3 Differentiation**[20 marks]**

- (a) Find the derivative of each function below. State which rule you are using in each case.

(i) $h(x) = 4x^3 - 2x^2 + x - 6$ *(3 marks)*

(ii) $k(x) = (x + 3)(x^2 - 2)$ Expand first, then differentiate. *(4 marks)*

(iii) $m(x) = (2x - 5)^3$ *(3 marks)*

- (b) A drone rises straight up from the ground. Its height in metres at time t seconds is given by:

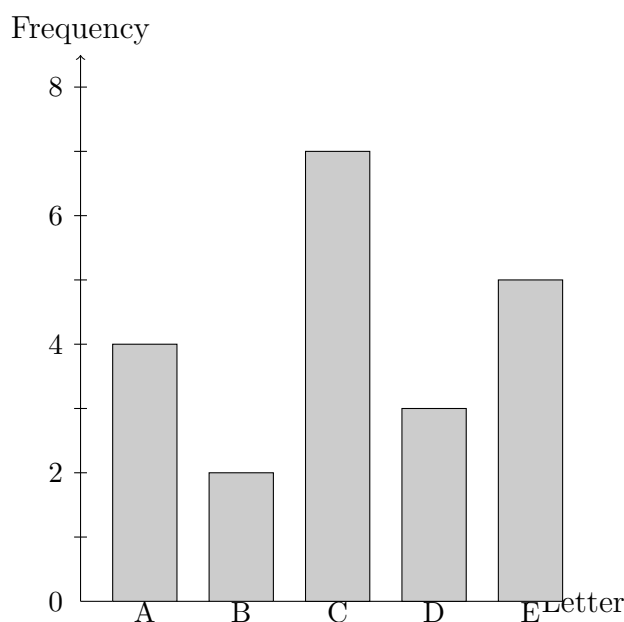
$$h(t) = 20t - 4t^2, \quad t \geq 0$$

- (i) Find $h'(t)$. In one sentence, describe what $h'(t)$ represents in this situation. *(3 marks)*
- (ii) At what time t does the drone reach its maximum height? *(2 marks)*
- (iii) What is that maximum height? *(2 marks)*
- (iv) After reaching its peak, the drone falls back to the ground. At what time t does it land? *(3 marks)*

Marks this page	
(a)(i)	/3
(a)(ii)	/4
(a)(iii)	/3
(b)(i)	/3
(b)(ii)	/2
(b)(iii)	/2
(b)(iv)	/3
Page total	/20

Question B4 Probability and Statistics**[20 marks]**

- (a) **Coin flips.** There is something genuinely surprising about what happens with long sequences of coin flips.
- (i) What is the probability of getting heads on a single flip? *(1 mark)*
 - (ii) What is the probability of getting heads **three times in a row**? Show your reasoning. *(2 marks)*
 - (iii) What is the probability of getting heads **eight times in a row**? *(1 mark)*
 - (iv) Suppose you flip a coin 1000 times. At each position from flip 1 through to flip 993, you can ask: “do the next 8 flips all come up heads?” That gives 993 possible starting positions for a run of 8 heads. Using your answer to (iii), estimate the expected number of 8-heads-in-a-row runs in 1000 flips. Round to one decimal place. Does the answer surprise you? *(4 marks)*
- (b) **Reading a chart.** The bar chart below shows how many times each of the letters A through E appeared in a short message.



- (i) Which letter appeared most often? How many times did it appear? *(1 mark)*
- (ii) How many letters are there in the message altogether? *(1 mark)*
- (iii) If one letter is chosen at random from the message, what is the probability it is the letter C? *(3 marks)*
- (iv) What is the probability the chosen letter is **not** C? *(1 mark)*
- (v) Find the mean frequency across the five letters A–E. *(3 marks)*
- (vi) List the five letters in order from **least likely** to **most likely** to be chosen at random. *(3 marks)*

Marks this page	
(a)(i)	/1
(a)(ii)	/2
(a)(iii)	/1
(a)(iv)	/4
(b)(i)	/1
(b)(ii)	/1
(b)(iii)	/3
(b)(iv)	/1
(b)(v)	/3
(b)(vi)	/3
Page total	/20

Question B5 Triangle Trigonometry**[20 marks]**

In triangle ABC : angle $C = 55^\circ$, side $a = BC = 6$ cm, and side $b = AC = 9$ cm.

- (i) Draw the triangle and label all known sides and the known angle. *(3 marks)*
- (ii) Using the Cosine Rule, find the length of side $c = AB$. Round to 2 decimal places.
(6 marks)
- (iii) Using the Sine Rule, find angle A . Round to 1 decimal place. *(6 marks)*
- (iv) Find angle B without any further calculation. *(2 marks)*
- (v) Calculate the area of triangle ABC using $\text{Area} = \frac{1}{2} ab \sin C$. *(3 marks)*

Marks this page	
(i)	/3
(ii)	/6
(iii)	/6
(iv)	/2
(v)	/3
Page total	/20

Question B6 Similar Triangles**[20 marks]**

Two right-angled triangles are given. In each triangle, the two shorter sides are **equal in length**. The longest side of the small triangle is 10 cm, and the longest side of the large triangle is 30 cm.

- (a) Draw both triangles side by side. For each triangle, mark the right angle, indicate that the two shorter sides are equal, and label the longest side with its length. Mark any angles you do not yet know with a letter or a question mark. *(4 marks)*
- (b) Using the fact that the angles of any triangle sum to 180° , find the size of each unknown angle in the small triangle. Explain your reasoning in full. *(4 marks)*
- (c) Find the length of each shorter side of the small triangle. Leave your answer in surd form. *(4 marks)*
- (d) Write down all three side lengths of the large triangle, and state the scale factor from the small triangle to the large one. *(4 marks)*
- (e) Calculate the area of each triangle, showing your working. By what factor is the area of the large triangle greater than the area of the small one? *(4 marks)*

Marks this page	
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(b)	/4
(c)	/4
(d)	/4
(e)	/4
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— END OF EXAMINATION —